



Genome-wide identification of *Aux/IAA* and *ARF* gene families in bread wheat (*Triticum aestivum* L.)

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Abstract

Wheat (*Triticum aestivum* L.) is one of the most important food crops in the world. Somatic embryogenesis is an event that is triggered by the presence of auxin hormone for the induction of somatic cells to get converted to embryonic cells. Somatic embryogenesis represents the most important process of totipotency of plants. The role of auxins is widely understood during various stages of embryogenesis including polarity establishment, de-differentiation, re-differentiations, and morphogenesis. Many of the *Aux/IAAs* and *ARFs* which are part of auxin signaling components have been identified to play various roles during embryogenesis. In this analysis, the *Aux/IAAs* and *ARFs* of *T. aestivum* have been analyzed at the genome-scale; their structure, function, and evolutionary relatedness were determined. Several *Aux/IAAs* and *ARFs* components of *T. aestivum* have been found to exclusively regulate axis formation, meristem commitment, and other re-differentiation processes by differential expression studies.

Keywords *T. aestivum* · *Aux/IAA* · *ARF* · Gene family · Phylogenetic analysis

Introduction

Auxin is the key hormone involved in the plant growth and development processes such as apical dominance, elongation of petiole and stem, roots architecture and hair formation, gravitropism, flower formation, vascular pattern development and fruit ripening via the interplay of *Aux/IAAs* (auxin/indole acetic acid) and *ARFs* (auxin response factor) genes/proteins (Kieffer et al. 2010; Kazan 2013; Schaller et al. 2015; Wang et al. 2010; Reed et al. 2001; Smet et al. 2007 and Woodward et al. 2005). *Aux/IAAs* belong to primary early auxin-response genes which include genes such as *GH3* (Gretchen Hagen 3) and *SAUR* (small auxin up RNA) (Shi et al. 2020). Auxin treatment induces the transcription of these sets of genes which are the key players at the early stage of auxin signal transduction.

Aux/IAA proteins consist of four highly conserved domains (domain I–IV). N-terminal consists of domain I

whereas C-terminal consists of domains III and IV of *Aux/IAA* proteins (Qiao et al. 2015). Domain I act as a transcriptional repressor (Tiwari et al. 2003) and domain III–IV mediates the polymerization of *Aux/IAA* and *ARF*, which eventually inhibits the *ARF* function (Maraschin et al. 2009; Piya et al. 2014). Domain II regulates the stability of *Aux/IAA* protein via ubiquitination complex *TIR1* (transport inhibitor response 1) (Kepinski and Leyser 2004, 2005; Hagen and Guilfoyle 2002). It consists of 13 amino acids which help in the formation of degron. Downstream to the auxin signalling pathway, *ARFs* are known to be involved in the *TIR1*/Auxin Signaling F-box (*TIR1/AFB*) signal cascade (Boer et al. 2014; Dinesh et al. 2015). The gene structure of *ARFs* generally consists of an amino-terminal DNA binding domain (DBD), a middle region (MR) that functions as either an activation domain or a repression domain, and a carboxy-terminal dimerization domain (CTD) which initiates the transport inhibitor response (Boer et al. 2014; Dinesh et al. 2015; Qiao et al. 2018). *Aux/IAAs* in the absence of auxin inhibit the function of *ARF* transcription factors by binding to the CTD domain of *ARFs* whereas the high auxin concentration promotes the *TIR1/AFB* mediated ubiquitin–proteasome-dependent degradation of *Aux/IAAs* freeing *ARFs* from the repression and carrying out the activation or repression of auxin-responsive genes (Jing et al.

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2015). The activation or repression occurs via the DBD that binds to auxin response elements (AuxREs) within the promoters of target genes (Dinesh et al. 2015; Winkler et al. 2017) (Fig. 1). *PIN-FORMED* (PIN) transporter proteins mediate the transport of auxin during organogenesis and pathogenesis in the plant (Kazan and Lynos 2014; Grones and Frimi 2015). *PIN1* has been shown to maintain a strong auxin concentration in the embryo and is regulated by *ARFs*. *Arabidopsis Aux/IAAs IAA20* and *IAA 30* have been shown to regulate root cap initiation in embryos (Sato and Yamamoto 2008), while *IAA9* from *Elaeis guineensis* is a predictable marker gene for embryogenesis establishment (Ooi et al. 2012). *Aux/IAAs* and *ARFs* play a crucial role as signaling genes in the eventful organization of somatic embryogenesis. Mahalakshmi et al. (2003) reported induction of somatic embryos in *T. aestivum* leaf base tissue treated with auxin for 24 h and the direct involvement of calcium ions (Mahalakshmi et al. 2007) in somatic embryogenesis. An interplay across auxin-induced somatic embryos, calcium ions and brassinosteroid via *Aux/IAAs* was reported (Singla et al. 2007).

In this study, we identified *Aux/IAA* and *ARF* genes in the allohexaploid *Triticum aestivum* genome. The role of various *Aux/IAAs* and *ARFs* in plant development and their comprehensive analysis could help in unravelling the agricultural traits associated with auxin in reverse genetics. The release of *Triticum aestivum* genomic data (International Wheat Genome Sequencing Consortium 2018) enables us to isolate the *Aux/IAA* and *ARF* gene family in *T. aestivum*. We, therefore, performed a detailed analysis of sequence alignment, phylogenetic relationship, gene structure, conserved domains, gene ontology enrichment, and network analysis. We hypothesize that this information would add

substantially to the functional characterization of *TaIAA* and *TaARF* involved in *Triticum aestivum* development at various stages.

Material and methods

Genome-wide identification of *Aux/IAAs* and *ARF* genes in *Triticum aestivum* L.

Aux/IAA and *ARF* genes from *T. aestivum* and two monocot species which include (*Hordeum vulgare*), rice (*Oryza sativa*), and a eudicot *Arabidopsis thaliana* were identified. Their protein sequences were obtained from ensembl genomes database (<ftp://ftp.ensemblgenomes.org/pub/plants/release50/>) and the HMM (hidden Markov model) profile was constructed from known *Aux/IAAs* (Pfam accession number: PF02309.11) and *ARFs* (Pfam accession number: PF06507.8) using *hmmsearch* program (HMMER-3.3.2; <http://hmmer.org/>). All identified *Aux/IAAs* and *ARFs* having e-values less than or equal to $1E-5$ were considered for downstream analysis.

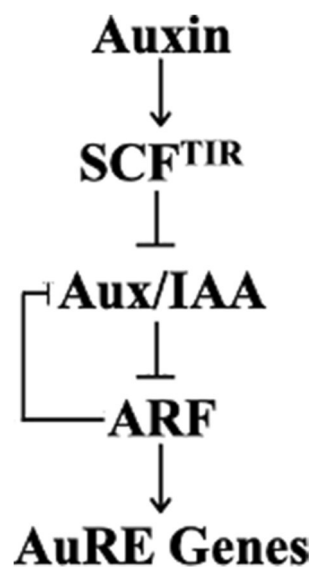
Primary sequence analysis of *T. aestivum* *ARFs* and *AUX/IAAs*

To draw the intron–exon junctions, coordinates of the genic components of the identified members were fetched from the GFF files (ftp://ftp.ensemblgenomes.org/pub/plants/release50/gff3/triticum_aestivum/). These coordinates were used on Gene Structure Display Server (GSDS) version 2.0 (<http://gsds.gao-lab.org/>; Hu et al. 2015) with coding sequences (CDS) and genomic sequences to generate the gene diagram displaying genic features including exons, introns, 5' and 3' UTRs. The basic physical and chemical parameters such as isoelectric point (pI) and molecular weight (MW) of *TaIAAs* and *TaARFs* were predicted using the Compute pI/MW tool (<https://www.expasy.org/resources/compute-pi-mw>; Bjellqvist et al. 1994) on ExPASy web server. MEME (Multiple Em for Motif Elicitation version 5.3.3) suit (<http://memesuite.org>; Bailey et al. 2009) was used to identify the conserved motifs in the *TaIAA* and *TaARF* protein sequences with the parameters configured to “Number of Motifs = 15” and “Optimum Motif Width = 8 to 50” residues. Subsequently, annotation of the identified motifs was done using InterProScan database with default parameters (<http://www.ebi.ac.uk/Tools/pfa/iprscan/>).

Phylogenetic analysis of *Aux/IAAs* and *ARFs* genes

The *IAA* and *ARF* protein sequences of *T. aestivum*, *H. vulgare*, *O. sativa*, and *Arabidopsis* were downloaded from the ensemble plants (<https://plants.ensembl.org/index.html>) and

Fig. 1 Auxin signaling pathway. The *Aux/IAA* gene participate in the auxin signaling pathway. *SCF^{TIR}* auxin transport inhibitor response 1-SKP1-Cullin-F-box complex, *ARF* auxin response factor, *AuxRE* auxin response genes



alignment was performed using Clustal omega web server (<https://www.ebi.ac.uk/Tools/msa/clustalo/>) with default parameters. A phylogenetic tree was constructed with the maximum likelihood (ML) method (1000-replicates bootstrap) using Molecular Evolutionary Genetics Analysis (MEGA X version-10.2.4) tool. An interactive tree of life (ITOL) webserver (<https://itol.embl.de/upload.cgi>) was used to visualize and edit the phylogenetic trees.

Gene ontology (GO) enrichment and network analysis

Gene ontology terms for identified members were fetched from the Gramene (ftp://ftp.gramene.org/pub/gramene/CURRENT_RELEASE/data/ontology/go/go_ensembl_triticum_aestivum.gaf) and an enrichment analysis (Hypergeometric test) with Benjamini and Hochberg False Discovery Rate (FDR) correction was performed using BiNGO plugin (<https://apps.cytoscape.org/apps/bingo>; Maere et al. 2005) in Cytoscape version 3.02 (Shannon et al. 2003; <https://cytoscape.org/download.html>). Scatterplots were constructed for the enriched GO terms using REVIGO (reduce + visualize gene ontology; <http://revigo.irb.hr/>) web server and network diagrams were plotted in cytoscape.

Plant material and stress treatments

T. aestivum cv. PBW343 seeds were thoroughly surface-sterilized for 20 min with 4% sodium hypochlorite solution and then were inoculated on MS basal media under the photoperiodic regimen of 16/8 h light and dark cycle for 10–13 days. Leaf base segments were harvested (2nd/3rd leaf) from the seedlings and then transferred to MS basal medium containing (i) Control medium (MS only) (ii) supplemented with 100 μ M 2,4-D, (iii) 320 μ M CaCl₂, and (iv) 10 nM epibrassinolide, for 24 h and then transferred to MS basal media for 13 days in the dark at 21 $\pm$ 2 °C respectively. The 2,4-D treated leaf bases developed somatic embryos on their tip after 2 weeks on MS basal medium. The treated tissues were then frozen with liquid nitrogen and stored at – 80 °C till further use.

Quantitative real time (qRT) PCR analysis

Total RNA from the *Triticum aestivum* leaf base was isolated using the TRIZOL reagent (Invitrogen) method and purified with Plant Mini elute Cleanup Kit (Qiagen, USA). cDNA was synthesized from 1 μ g RNA using Verso cDNA Synthesis Kit (ThermoFisherScientific, USA), according to the manufacturer's instructions. qPCR analysis was performed on Mx3005P (Agilent Technologies) system using SYBR Green Brilliant III Ultra-Fast (Agilent Technologies). Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) was

used as an internal control for the analysis (Jarosova and Kundu, 2010). Amplification was carried out at 95 °C for 2 min; 40 cycles of 95 °C at 15 s; 60 °C for 1 min followed by melting curve analysis to ensure that each amplicon was a single product. Two biological replicates and their 3 technical replicates were used in real-time PCR analysis. The 2– $\Delta\Delta$ ct method was used for determining the calculation of fold-change in activity.

Results

Identification and structural features of *Aux/IAA* and *ARF* genes

Aux/IAA and *ARF* genes were identified using hmm search in *T. aestivum*. A total of 84 *Aux/IAA*s and 67 *ARF*s members were identified from the ensemble build-50 of the *T. aestivum* genome (Supplementary Tables 1 and 2). The average gene length of *TaAux/IAA*s and *TaARF*s were found to be approximately 1.1 kb and 3 kb long, respectively. The minimum gene length was observed of 402 bp for the *Ta5D-IAA26* gene and 1125 bp for *Ta7A-ARFM*, whereas *Ta3D-IAA5* and *Ta7A-ARFS* encode the largest genes of 3.5 kb and 5 kb, respectively (Fig. 2A, B). Twelve *TaAux/IAA*s and four *TaARF*s members did not have a UTR region in their cDNA. The exon–intron structures of *T. aestivum* *Aux/IAA* and *ARF* gene family were analyzed using Gene Structure Display Server (GSDS). *TaIAA* and *TaARF* genes were interrupted by 1–5 and 1–14 introns, respectively. There were no significant differences in the number of exons among members of the same clade, but noteworthy differences existed among sister pairs (Fig. 2A, B).

Characteristics of *TaAux/IAA* and *TaARF* protein sequences

The basic information containing gene name, locus, protein length, intron number, predicted isoelectric focusing points (PI), and molecular weight were identified (Supplementary Tables 1 and 2). *TaAux/IAA*s and *TaARF*s have shown a broader range of PI, suggesting their wide range of activities. The minimum PI measured for the *TaAux/IAA*s and *TaARF*s were 4.0 (*TaIAA29*) and 5.47 (*TaARFM*), respectively, whereas the highest PI measured was 9.19 (*TaIAA27*) and 8.16 (*TaARFC*). The molecular masses of the identified proteins also varied considerably. Among the *TaAux/IAA*s, *TaIAA27* encodes the largest protein of 36.9 KDa while *TaIAA26* encodes the smallest protein of molecular mass of 15.43 KDa whereas, the *TaARFS* and *TaARFM* encode 128.28 KDa of large protein and 57.22 KDa of smallest protein among the *TaARF*s.

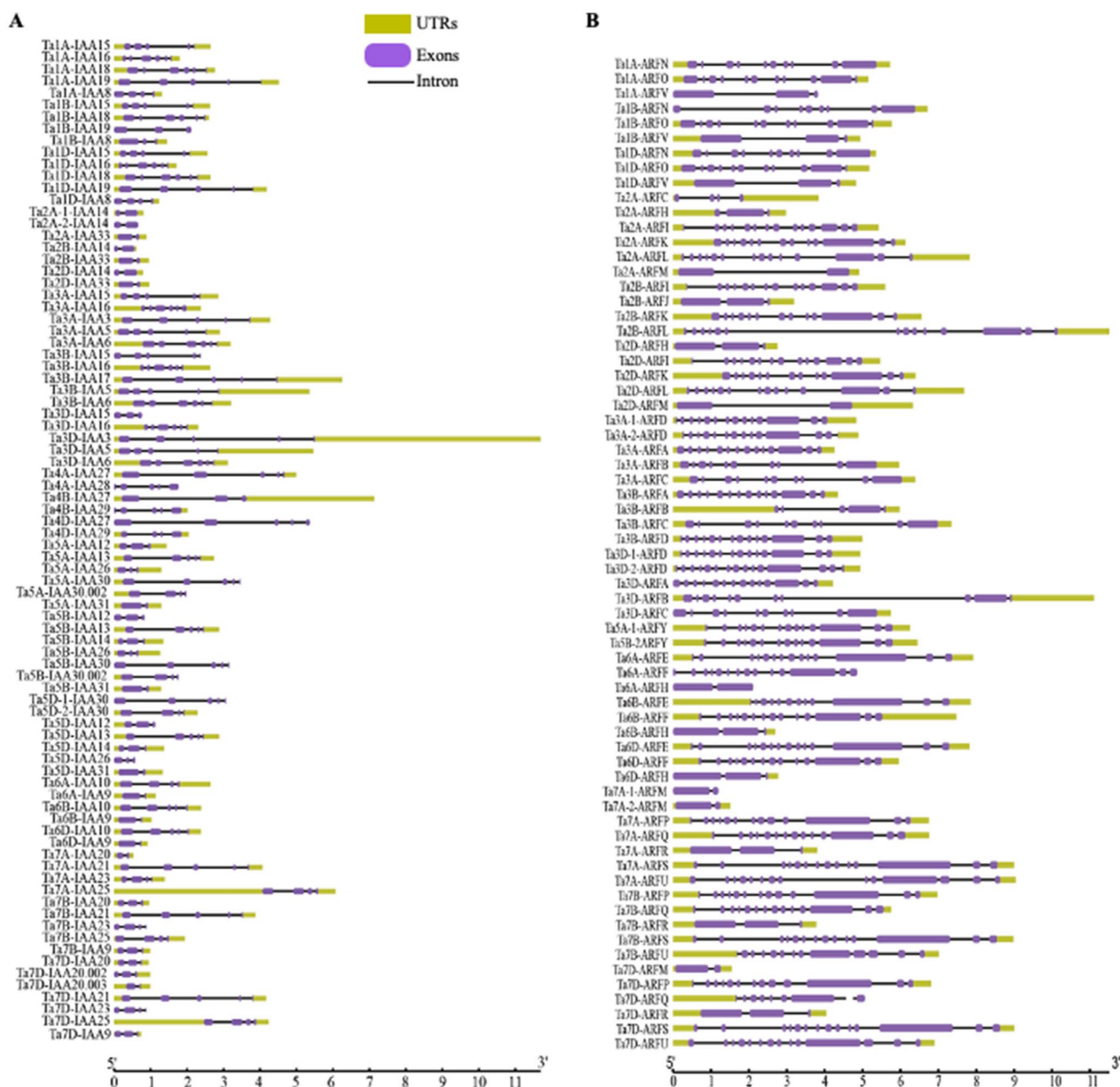
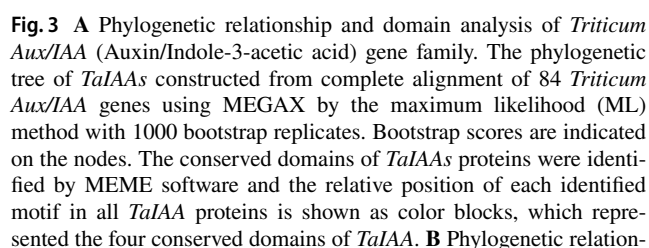


Fig. 2 Gene structure analysis. Illustration depicting the distribution of exon, intron and UTRs (5' and 3') for (A) **Aux/IAA**, (B) **ARF** gene family members of *Triticum aestivum*. Purple and black lines represent exons and intron respectively whereas green boxes represent UTRs

Phylogenetic relationship and motif analysis of TaAux/IAA and TaARF members

All the members of the Aux/IAA and ARF gene family showed a strong bootstrap replication support (> 800) (Fig. 3A, B) along with the distribution of members across 5 (Aux/IAA) and 6 (ARF)/clades. Identified 84 TaAux/IAA members are distributed into five clades harbouring 9 members in clade I, 16 members in clade II, 4 members in clade III, 14 members in clade IV, and 41 members in clade V (Fig. 3A), whereas for the TaARF gene family we

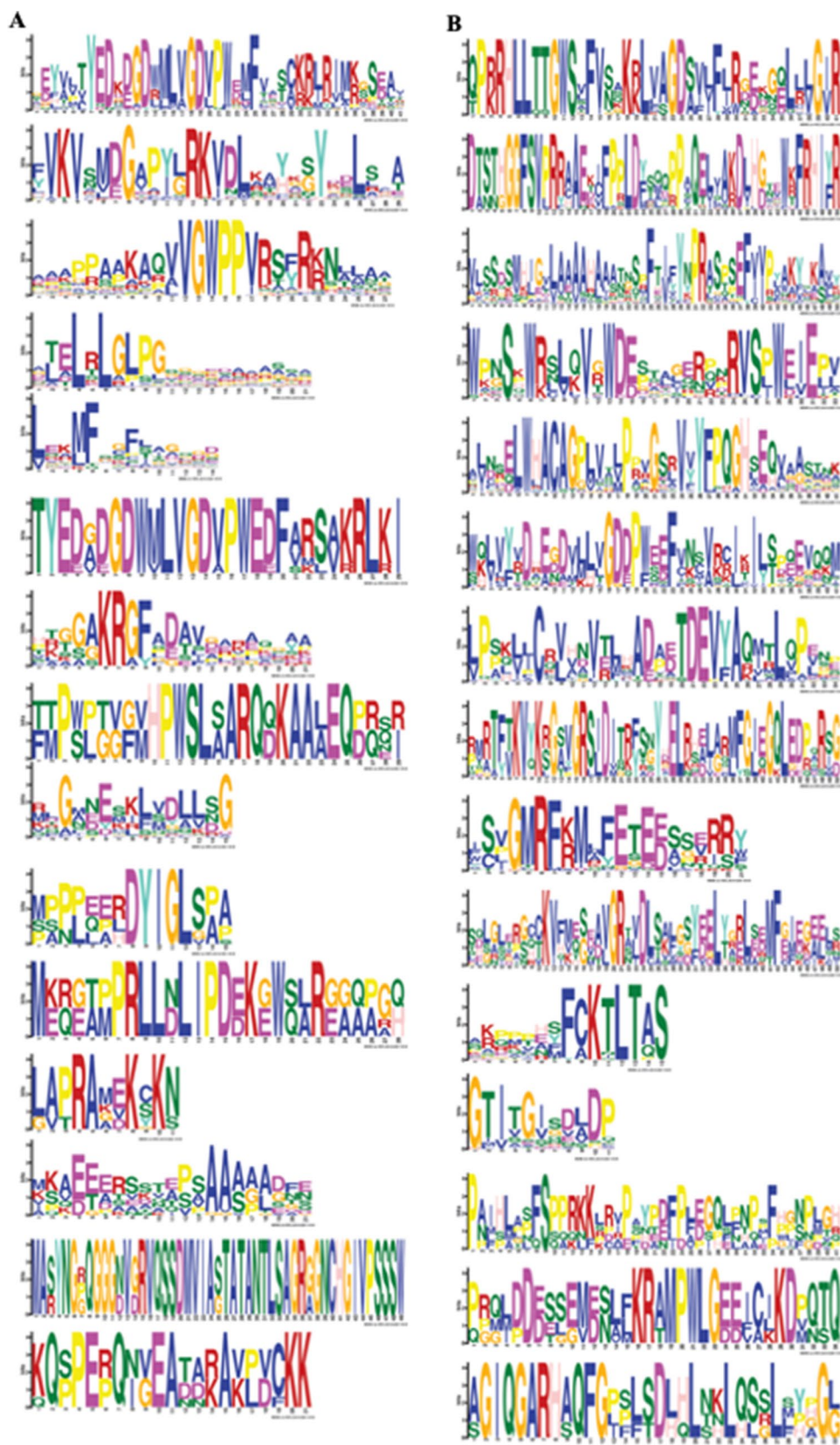
have identified 67 members distributed across six clades harbouring 6 members in clade I, 2 members in clade II, 3 members in clade III, 12 members in clade IV, 14 members in clade V, and 30 members in clade VI (Fig. 3B). The motif detection software MEME was used to perform motif analysis of the Aux/IAA and ARF gene family. The number of motifs contained in Aux/IAs and ARFs ranged from 8 to 13 (Fig. 4A, B) and the distribution of the motifs was relatively conservative among the internal members of the clade.



Phylogenetic relationship among the Aux/IAA and ARF proteins of *T. aestivum* and relative species such as *Hordeum vulgare* (33 Aux/IAs and 28 ARFs), *Oryza sativa* (35 Aux/IAs and 28 ARFs), and *Arabidopsis thaliana* (28 Aux/IAs and 22 ARFs), were performed (Fig. 5A, B). Aux/IAA and ARF proteins were classified into seven (I to VII)

and six (I to VI) major groups, respectively. Group VI was the largest of all groups encompassing IAAs from all the species (Fig. 5A). The TaIAA33 located on chromosome 2 showed similarity to Hv1-IAA33 and OsIAA33 whereas TaIAA10 and TaIAA9 located on chromosome 6 were distributed along with the *H. vulgare* and *O. sativa* IAAs (Hv6-IAA10, Hv6-IAA9, OsIAA10, OsIAA9). Ta1-IAA8 was closely related to the *H. vulgare* and *O. sativa* IAA8 and formed the same clade whereas TaIAA20, TaIAA9,

Fig. 4 Motif analysis. Illustration depicting the distribution of motifs for **a** *TaAux/IAA* and **b** *Ta Auxin response factor* (ARF) gene family members of *Triticum aestivum*. Motifs are identified using MEME package



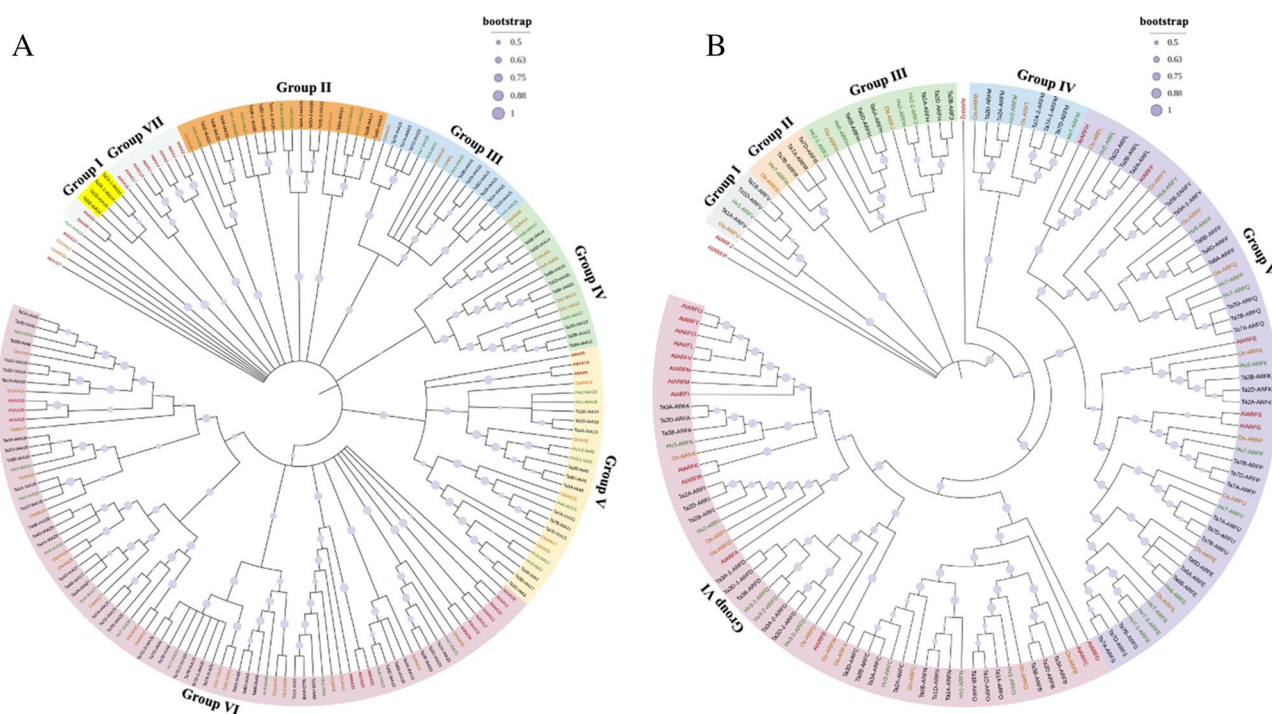


Fig. 5 **A** Phylogeny analysis. Phylogenetic tree depicting the evolution of *Aux/IAA* gene family members in *Triticum*. The full-length amino acid sequences of *Triticum aestivum*, *Hordeum vulgare*, *Oryza sativa*, and *Arabidopsis* *Aux/IAA* proteins are aligned using Clustal omega and the maximum likelihood (ML) tree is constructed using MEGAX with 1000 bootstrap replicates. Seven groups are highlighted in different color codes. *Triticum*, *H. vulgare*, *O. sativa* and *Arabidopsis* *Aux/IAAs* are represented by black, green, orange, and red color respectively. Purple circles on the node of the tree shows the bootstrap replication values obtained after 1000 replication. **B**

Phylogeny analysis. Phylogenetic tree depicting the evolution of *ARF* gene family member in *Triticum*. The full-length amino acid sequences of *Triticum*, *H. vulgare*, *O. sativa*, and *Arabidopsis* *Aux/IAA* proteins are aligned using Clustal Omega and the maximum likelihood (ML) tree is constructed using MEGAX with 1000 bootstrap replicates. Seven groups are highlighted in different color codes. *Triticum*, *H. vulgare*, *O. sativa*, and *Arabidopsis* *ARFs* are represented by black, green, orange and red color respectively. Purple circles on the node of the tree shows the bootstrap replication values obtained after 1000 replication

and TaIAA25 were located on chromosome 7 and showed similarity to *H. vulgare* and *O. sativa* IAA (Hv7-IAA20, Hv7-IAA25, OsIAA20, OsIAA22, OsIAA25). TaIAA27, TaIAA28, TaIAA29 were distributed along chromosome 4 showing similarities to *H. vulgare* and *O. sativa*. TaIAA18 and TaIAA16 were located on chromosome 1, and interestingly OsIAA2 showed more similarity with Ta-IAA16.

In group I, Ta2-IAA14 showed similarity to Arabidopsis IAA (i.e., AtIAA2, AtIAA1, AtIAA3, AtIAA4, AtIAA7, AtIAA14, AtIAA17, AtIAA16, AtIAA9, and AtIAA9), whereas, group II TaIAAs (TaIAA26, TaIAA30, TaIAA13) were closely related to *H. vulgare* (Hv5-IAA26, Hv5-IAA30, Hv5-IAA17, Hv5-IAA30, Hv5-IAA11, Hv5-IAA13) and *O. sativa* (OsIAA26, OsIAA13) IAA. Group III has Ta7-IAA23, Ta3-IAA15, Ta1-IAA15, and these three members were closely related to Hv7-IAA23, Hv3-IAA1, Hv3-IAA15 of *H. vulgare* and OsIAA23, OsIAA15, OsIAA1 of *O. sativa*.

Group IV TaIAAs (TaIAA14, TaIAA31, TaIAA12) formed the clade with *H. vulgare* (HvIAA14, HvIAA12) and *O. sativa* (OsIAA24, OsIAA14, OsIAA31, OsIAA12)

whereas, group V showed similarity across all the species, i.e., *T. aestivum* chromosome 1,3, and 7 (TaIAA19, TaIAA5, TaIAA21, TaIAA3, TaIAA7, TaIAA13), *H. vulgare* (HvIAA19, HvIAA5, HvIAA21, HvIAA17, HvIAA3), *O. sativa* (OsIAA19, OsIAA5, OsIAA21, OsIAA17, OsIAA3), and Arabidopsis (AtIAA5, AtIAA19, AtIAA6) (Fig. 5A).

Contrary to the genes, ARF proteins were distributed into six groups (group I to group VI). Group I included the ARFV proteins whereas group II formed a small group of ARFR proteins from *T. aestivum*, *H. vulgare*, and *O. sativa*. Group III and IV represented the ARFH and ARFM proteins, respectively, across the three monocots (i.e., *T. aestivum*, *H. vulgare* and *O. sativa*). Group V and VI formed the largest cluster of ARF proteins in the phylogenetic analysis. Group V is enriched with the ARF proteins like ARFL, ARFY, ARFF, ARFQ, ARFK, ARFP, ARFU, ARFE, and ARFS. Following group V, group VI includes the proteins such as ARFB, ARFO, ARFN, ARFC, ARFD, ARFI, ARFA etc. (Fig. 5B). However, most of the Arabidopsis IAA and ARFs formed separate clades from *T. aestivum*, *H. vulgare*, and *O. sativa* which shows its distant relation to the monocots.

Gene ontology (GO) enrichment and pattern map of TaAux/IAAs and TaARFs

GO enrichment analysis for TaAux/IAA and TaARF members was performed using the “bingo” plugin of the Cytoscape tool and GO terms were functionally categorized into three ontological processes, i.e., biological processes (BP), cellular component (CC), and molecular function (MF). Under the BP category, TaAux/IAA proteins were involved mainly in auxin stimulus, organ development, root development, embryonic pattern specification, floral whorl development, lateral root morphogenesis, post-embryonic root morphogenesis, xylem and phloem pattern formation post-embryonic root development, and stamen filament development (Fig. 6A), whereas TaARF proteins were involved in postembryonic development, post-embryonic organ development, anatomical

structure development, root system development, regulation of anthocyanin biosynthetic process, abaxial cell fate specification, pattern specification process, embryonic axis specification, phyllome development, and response to auxin stimulus (Fig. 6B).

The constructed pattern map using BP terms classified the TaAux/IAAs and TaARFs into three patterns, i.e., pattern A; biological regulation, pattern B; developmental process and pattern C; signalling (Fig. 7A) and TaARFs expressions were classified as pattern A; developmental process, pattern B; response to stimulus and pattern C; biological regulation (Fig. 7B). TaARFs regulate the processes related to the embryonic pattern, axis specification, lateral root formation, shoot development, ovule development, regulation of flower development, cell fate specification, response to auxin stimulus, auxin-mediated signalling pathway, and TaAux/IAAs showed the enrichment of

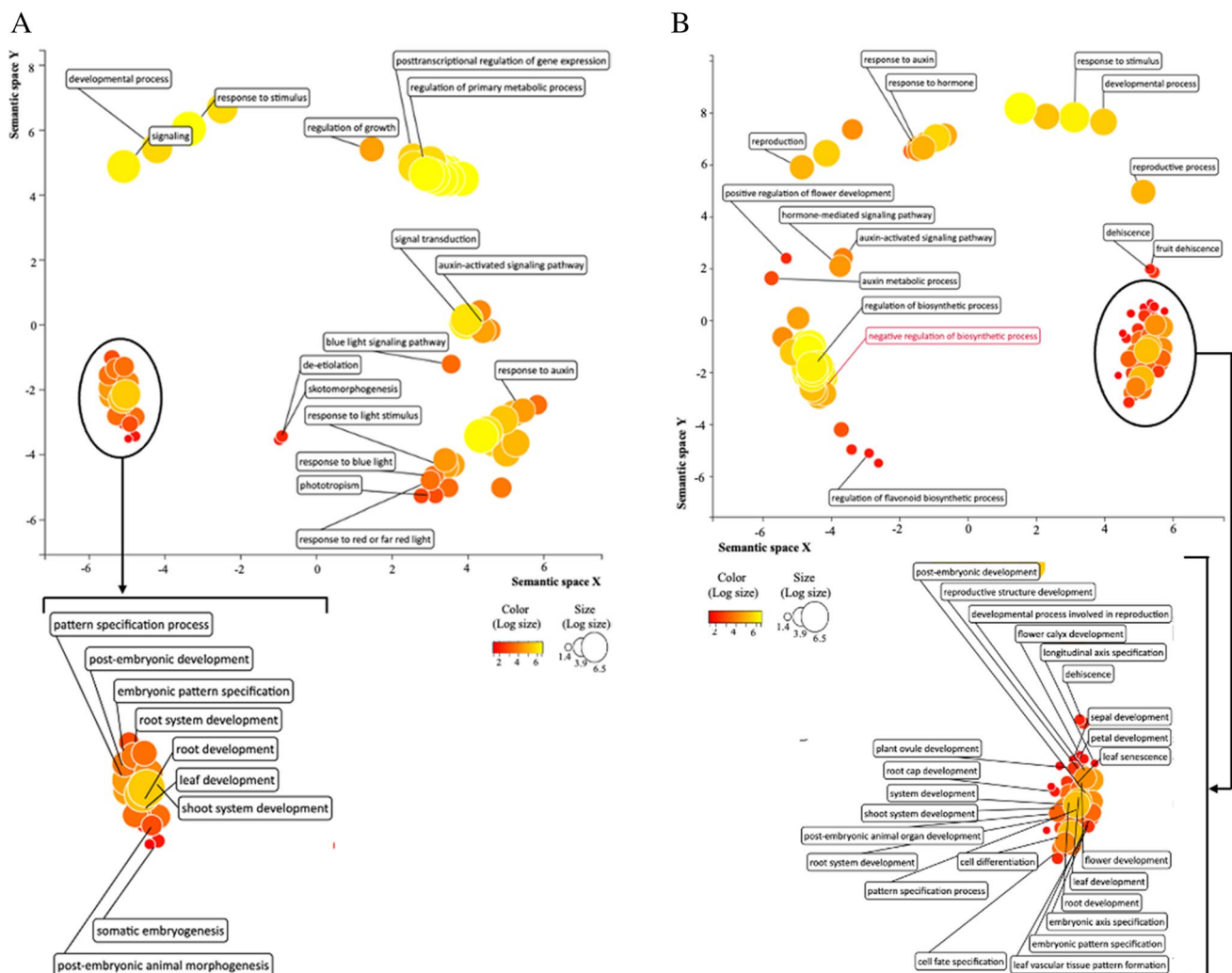


Fig. 6 **A** GO enrichment analysis. Scatterplot showing the enriched GO categories (biological processes) for *Aux-IAA* gene family in *Triticum* using REVIGO. **B** GO enrichment analysis. Scatterplot showing

the enriched GO categories (biological processes) for *ARF* gene family in *Triticum* using REVIGO

The diagram illustrates the interconnectedness of various biological processes, categorized into four main clusters:

- Metabolic processes:** This cluster includes processes such as "regulation of gene expression", "regulation of protein metabolic process", "regulation of primary metabolic process", "regulation of macromolecule metabolic process", "regulation of cellular protein metabolic process", "regulation of metabolic process", "regulation of biological process", "regulation of cellular metabolic process", "regulation of biosynthetic process", "regulation of cell cycle", "regulation of translation", and "posttranscriptional regulation of gene expression".
- Signaling cascade:** This cluster includes processes such as "auxin mediated signaling pathway", "hormone mediated signaling pathway", "cellular response to auxin stimulus", "response to auxin stimulus", "response to hormone stimulus", "response to ethylene stimulus", "response to endogenous stimulus", "response to abiotic stimulus", "intracellular signaling pathway", "intracellular receptor mediated signaling pathway", "cellular response to radiation", "cellular response to blue light", "cellular response to light stimulus", "response to light intensity", "response to absence of light", "response to red or far red light", "response to flight stimulus", "response to blue light", "response to radiation", "response to abiotic stimulus", "response to external stimulus", "response to stimulus", "signaling pathway", and "biological regulation".
- Development:** This cluster includes processes such as "cotyledon vascular tissue pattern formation", "xylem and phloem pattern formation", "regionalization", "stamen filament development", "leaf development", "shoot development", "post-embryonic organ development", "post-embryonic root development", "root development", "shoot system development", "root system development", "lateral root primordium development", "lateral root development", "root morphogenesis", "lateral root morphogenesis", "post-embryonic root development", "multicellular organismal development", "multicellular organismal process", "trapism", "phototropism", "gravitropism", "cellular response to blue light", "cellular response to light stimulus", "response to light intensity", "response to absence of light", "response to red or far red light", "response to flight stimulus", "response to blue light", "response to radiation", "response to abiotic stimulus", "response to external stimulus", "response to stimulus", "signaling pathway", and "biological regulation".
- Abiotic stimulus:** This cluster includes processes such as "response to blue light", "response to radiation", "response to abiotic stimulus", "response to external stimulus", "response to stimulus", "signaling pathway", and "biological regulation".

The diagram uses colored circles to represent different levels of process specificity: yellow circles represent specific processes, orange circles represent more general processes, and purple circles represent the most general processes. The connections between these circles show the complex interrelationships between these biological processes.

biological process Gene Ontologies term of biological process categories of *ARFs*. Node size represents the number of genes assigned to that particular GO term while colors represent the significance values (hypergeometric distribution test and p value ≤ 0.05)

is upregulated upon 2,4-D induction after 13 days. TaIAA15 was significantly upregulated in response to 2,4-D, CaCl₂, and EBL at 24 h whereas TaIAA14 was downregulated at 13 days. Auxin responsive proteins such as TaIAA18 showed a higher expression (8–9-folds) pattern in response to CaCl₂ and EBL treated leaf base at 24 h. However, its expression depleted in response to 2,4-D, CaCl₂, and EBL at 13-day treated leaf bases whereas TaIAA31 showed upregulation in response to CaCl₂ (~36-folds) and EBL (~19-folds) at 24 h (Fig. 8A–E).

TaARFs showed the highest expression ranging from ~2500 to ~4300-folds in 13-day-old 2,4-D treated leaf bases (somatic embryos; SEs) compared with control. CaCl₂

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B

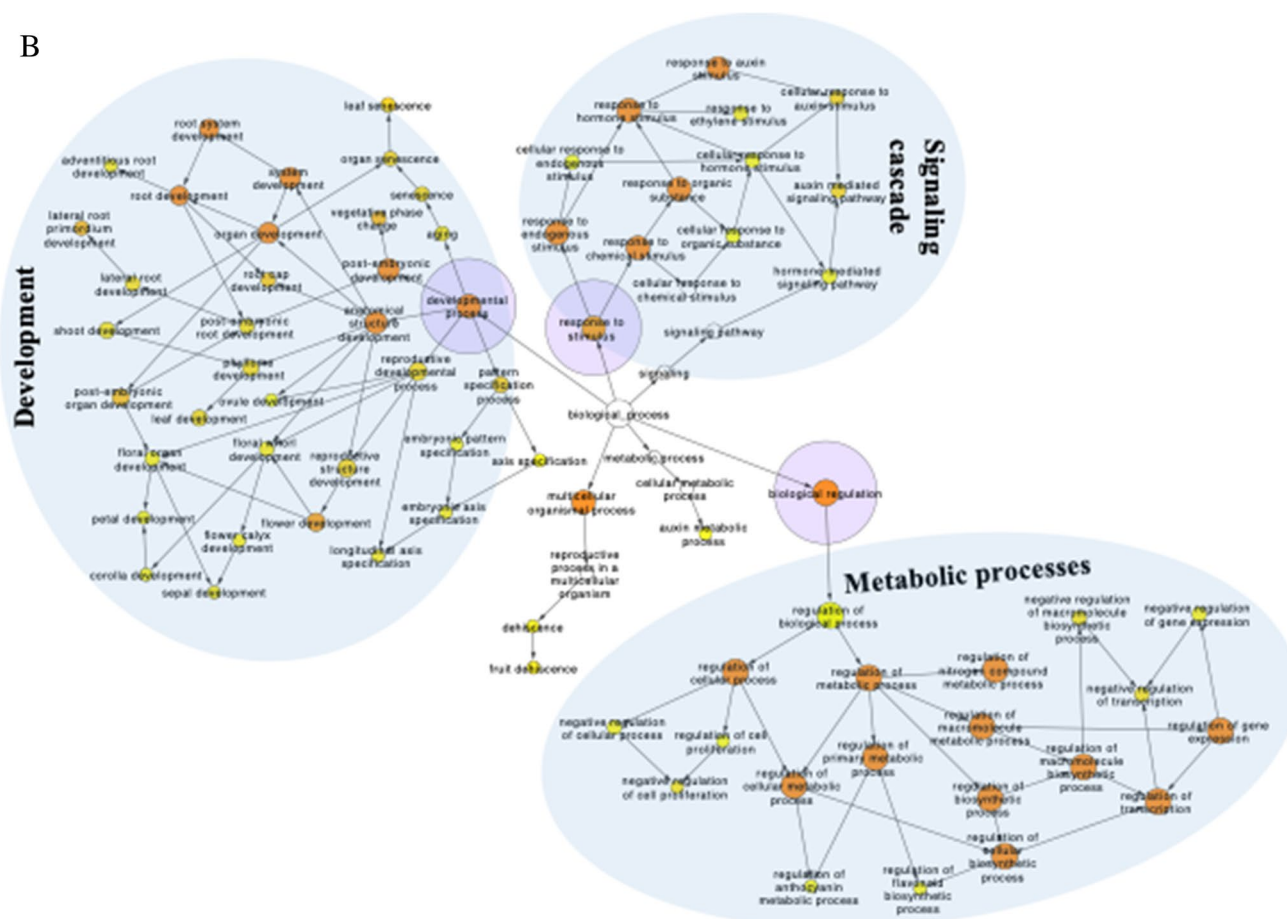


Fig. 7 (continued)

and EBL-treated tissues. Ta5B-ARFY, Ta2B-ARFL, Ta1B-ARFO, and Ta7B-ARFS showed significant upregulation (~15-folds) in SEs whereas these genes were downregulated in response to 2,4-D and CaCl₂ treatment at 24 h. ARFs such as Ta2A-ARFL, Ta7A-ARFP, and Ta2B-ARFL showed higher expression ranging from 4 to 5 folds in response to EBL treatment at 24 h (Fig. 9A–H).

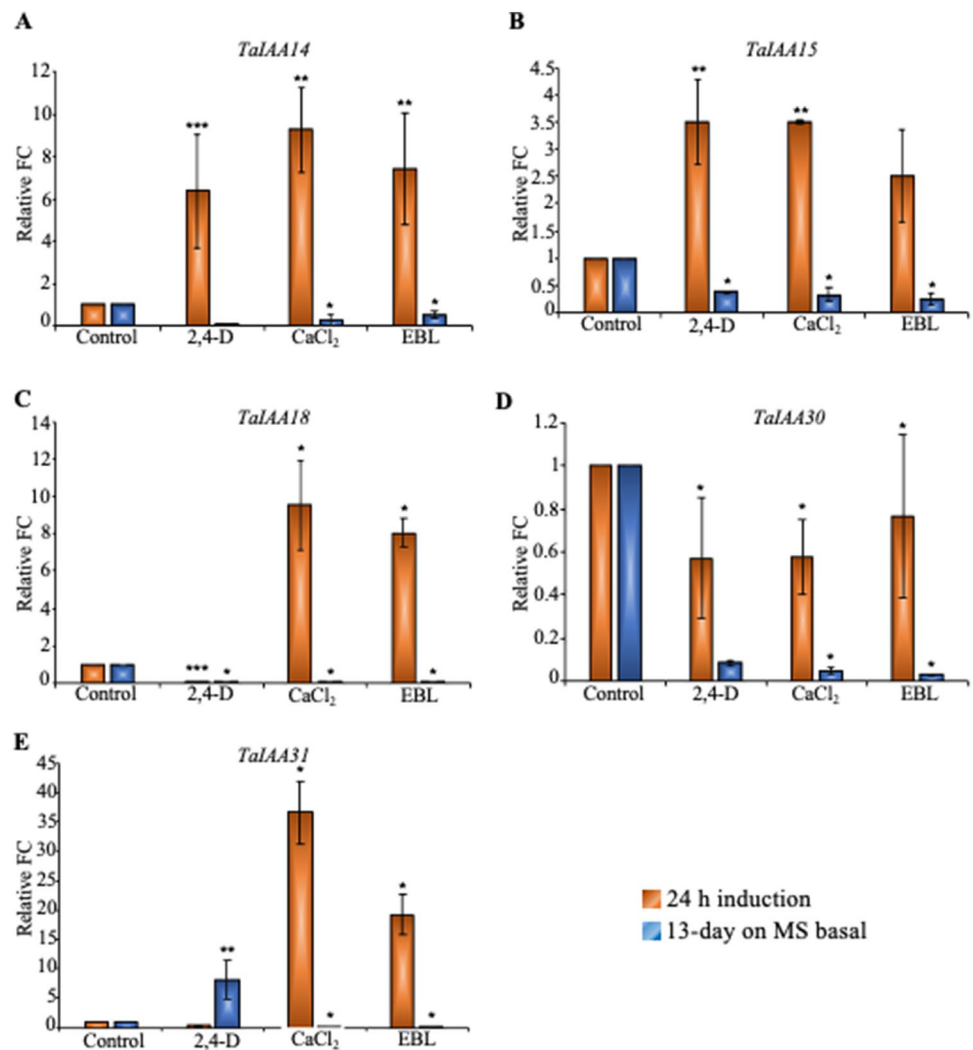
Discussion

The interplay of *Aux/IAA* and *ARF* genes in the auxin signaling pathway regulates the expression of auxin-responsive genes which plays an important role in plant development. As the high-resolution bread wheat genome is available, we extracted the *Aux/IAA* and *ARF* gene families from the hexaploid wheat. Previous studies reported that *Aux/IAA* genes from economic important crops such as tomato [26], rice [31], and maize [31] are similar to *Aux/IAA* genes from the wheat ancestral genomes, i.e., *T. urartu* [27] and *A. tauschii* [28] (Qiao et al. 2015). On the contrary, we have identified

high number of *Aux/IAA* family members in *T. aestivum* [84] probably because of the various allopolyploidization and duplication events among *T. turgidum* and *A. tauschii*. Moreover, neo-functionalization events were the distinctive feature of duplicated *Aux/IAA* genes (Yang et al.; 2006) which contributed to the alteration of conserved motifs. *ARF* genes from grass species such as barley [25], *Brachypodium distachyon* [28], and rice [25] (Wang et al.; 2007) are similar to *Aux/IAA* genes from the wheat ancestral genomes, i.e., *T. urartu* [21], *A. tauschii* [23], and *T. turgidum* [25] (Qiao et al. 2018). We have identified 67 *ARF* gene family members in *T. aestivum* which reflects its hexaploidy status.

The number of introns and exons varied greatly in both the gene families and *TaARF* members showed a large number of introns than *TaAux/IAA* members. *TaIAA* and *TaARF* genes were interrupted by 1–5 and 1–14 introns, respectively. The intron–exon structure analysis suggests an evolutionary conserved *IAA* and *ARF* gene family and physicochemical changes in the properties of amino acids indicate the functional divergence between clades and changes in the rate of evolution. Phylogenetic analysis revealed paralog

Fig. 8 qPCR of some *TaIAAs* under different conditions (i.e., 2,4-D, CaCl_2 , and EBL) at 24 h (orange bars) and 13-day-old (blue bars) *Triticum* leaf bases. **a** *TaIAA14*, **b** *TaIAA15*, **c** *TaIAA18*, **d** *TaIAA30*, and **e** *TaIAA31*



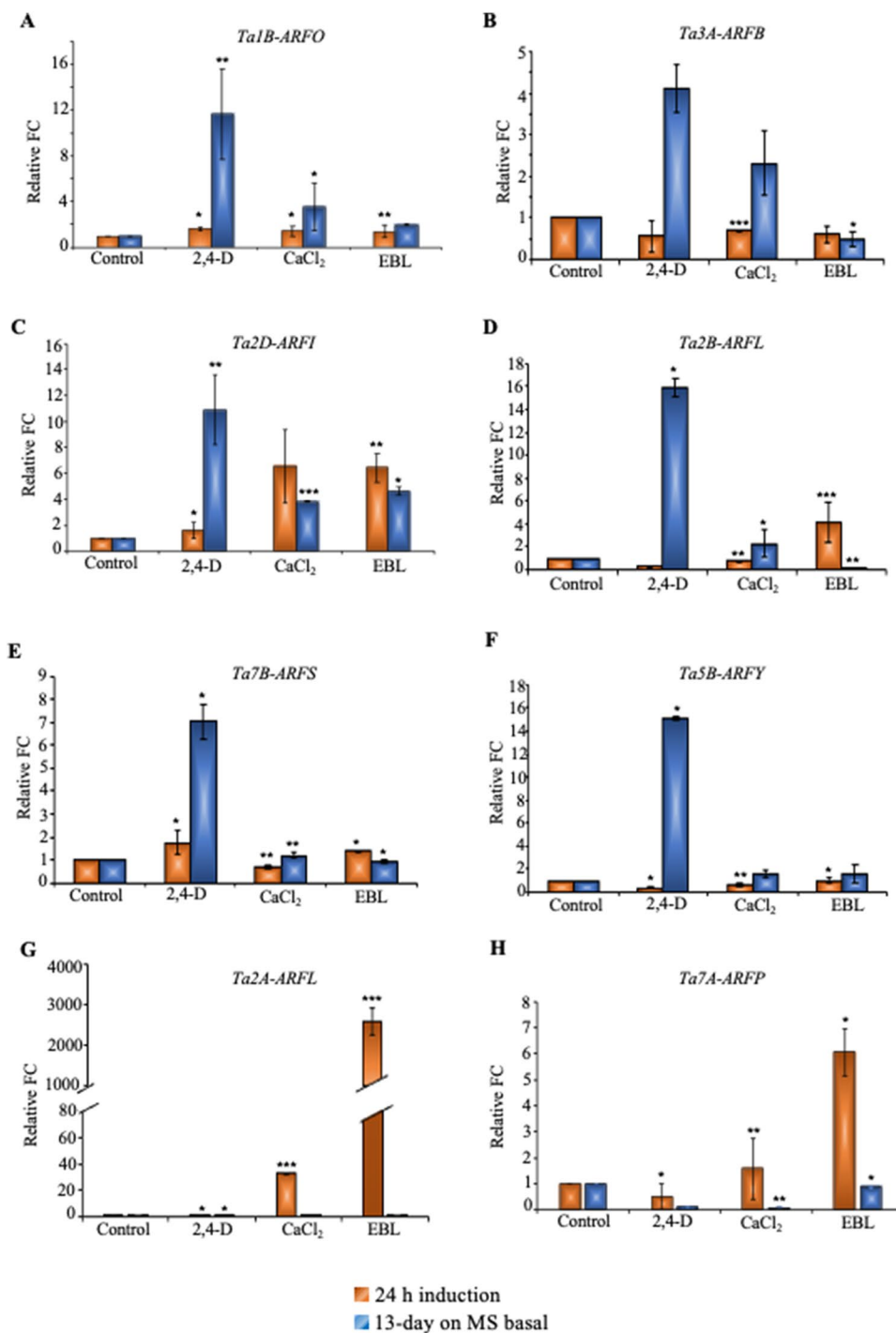
or ortholog pairs on the basis of topology distribution in accordance with the level of similarity among sister pairs present at the terminal nodes. Both the gene families were enriched for GO terms related to pattern specification processes, embryonic development, root-shoot development, somatic embryogenesis related genes, and skotomorphogenesis. Members of *TaAux/IAA* and *TaARF* gene families were found to be involved in the development, hormone signaling, and metabolic processes. We speculate that genes involved in polarity establishment and embryo pattern specification, which are the first committed steps towards embryo development are enriched significantly. Interplay across *Aux/IAA* and *ARF* genes is required in a spatio-temporal manner to establish polarity establishment. *TaAux/IAAs* such as *TaIAA14*, *TaIAA15*, and *TaIAA30* and *ARFs* like *ARFA*, *ARFC*, *ARFE*, *ARFF*, *ARFG*, and *ARFH* have been found to be involved in somatic embryogenesis. Recently, miRNA 160 mediated regulation of various embryonal developmental stages have been shown via *ARF10*, 16, and 17 activities (Lin et al. 2015).

Expression analysis of *TaAux/IAA* and *TaARF* genes in the leaf base tissue treated with the somatic embryogenesis inducing agents, i.e., 2,4-D, EBL and CaCl_2 were performed. The *TaAux/IAA* genes exhibited upregulation in EBL and CaCl_2 (24 h)-treated leaf bases, whereas *TaIAA14* and *TaIAA15* showed increased expression in 24 h 2,4-D treated samples and *TaIAA30* levels were down in all the treatments. On the contrary, *TaARF* genes were found to be highly expressed in somatic embryos in the 13 days after 24 h 2,4-D-treated samples. This study suggests that *Aux/IAAs* and *ARFs* regulate different stages of plant development such as somatic embryogenesis, root-shoot development, and pattern specification processes.

Conclusions

A significant progress has been made in the understanding of auxin perception and signaling transduction processes using various in silico, forward and reverse genetics approaches.

Fig. 9 qPCR of some *TaARFs* under different conditions (i.e., 2,4-D, CaCl_2 , and EBL) at 24 h (orange bars) and 13-day-old (blue bars) *Triticum* leaf bases. **a** *Ta1B-ARFO*, **b** *Ta3A-ARFB*, **c** *Ta2D-ARFI*, **d** *Ta2B-ARFL*, **e** *Ta7B-ARFS*, **f** *Ta5B-ARFY*, **g** *Ta2A-ARFL*, **h** *Ta7A-ARFP*



Crops such as wheat, rice, and barley are important worldwide and consist of complex genomes. In summary, 84 *Aux/IAAs* and 67 *ARFs* were identified from the bread wheat and both the gene families were found to be structurally highly conserved. In silico identification combined with genetic methods will accelerate the identification of genes involved in the regulation of *Aux/IAA* and *ARF* genes, which will eventually enhance the understanding of plant development in response to auxin.

Key message

Eighty-four *TaAux/IAAs* and 67 *TaARFs* were identified in *Triticum aestivum* and their structure, evolution, function was studied in order to unravel their role in somatic embryogenesis.

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Authors' contributions CC and NS performed the experiments. CC and PK wrote and formatted the manuscript. All authors read and approved the final manuscript.

Declarations

Conflicts of interest The authors declare that they do not have any conflict of interest.

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